Tcs AI Build Phase Problem Statements

Bachelors of Engineering - Computer Science (Chandigarh University)

tcs^{AI} Hackathon
Open Hack Problem Statements

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E: Low-code/No-code for non-coders!

Here are some of the problems (E1 to E5) curated for you to build applications leveraging free AI-powered no-code, low-code platforms with natural language prompts.

E1: RESTAURANT MENU OPTIMIZATION

Problem Description:
Restaurant chain with 25 locations loses $150K annually from poor menu decisions. No real-time insights on dish profitability, customer preferences, or seasonal demand. Management can't optimize pricing or identify underperforming items, leading to kitchen waste and declining satisfaction. You are required to build an application addressing restaurant's problem.

Expected Functionalities:
• Real-time dashboard showing menu performance, profit margins, popularity scores
• AI recommendations for menu changes and pricing adjustments
• Customer review sentiment analysis integration
• Seasonal demand forecasting
• Competitor price monitoring
• Kitchen waste optimization suggestions
• Mobile-responsive management interface
• POS system integration
Tips: Try AI-powered web application platform such as lovable

E2: SMALL BUSINESS FINANCIAL DASHBOARD

Problem Description:
Small business owners managing multiple revenue streams lack financial visibility and forecasting capabilities. 70% struggle with cash flow management, miss tax deadlines, and can't predict seasonal trends. Average business loses $50K annually due to poor financial planning and delayed decision-making.

Expected Functionalities:
• Real-time financial dashboard with revenue, expenses, and profit tracking
• Cash flow forecasting with seasonal trend analysis
• Automated expense categorization and tax preparation
• Invoice management with payment reminder automation
• Financial goal setting and progress tracking
• Integration with bank accounts and payment processors
• Customizable financial reports for stakeholders
• Alert system for budget overruns and payment due dates

Tips: Try AI-powered web application platform such as lovable

E3: MEDICAL RESEARCH ASSISTANT

Problem Description:
Hospital research department with 15 physicians spends 8+ hours weekly manually reviewing medical literature. This reduces patient care time by 20% and leads to missed breakthrough treatments, inconsistent protocols, and difficulty finding relevant research for patient cases.

Expected Functionalities:
• Upload and analyze medical papers/clinical trials for key insights
• Generate case-specific research summaries for patient conditions
• Compare treatment protocols with effectiveness ratings
• Set alerts for new research in medical specialties
• Create evidence-based treatment recommendations with citations
• Collaborative annotation system for physician teams
• Compliance checking against medical guidelines
• Automated literature monitoring and synthesis
Tips: Try AI-powered tool such as NotebookLM

E4: E-COMMERCE CUSTOMER SUPPORT INTELLIGENCE

Problem Description:
Online retailer with 50K monthly customers faces 40% cart abandonment and receives 500+ daily support queries. Customer service team can't identify purchase barriers, track satisfaction trends, or provide personalized assistance. Average response time is 12 hours, causing $200K monthly revenue loss.

Expected Functionalities:
• AI chatbot for instant customer query resolution
• Cart abandonment analysis with intervention triggers
• Customer behavior tracking and purchase prediction
• Automated support ticket categorization and routing
• Real-time satisfaction scoring and sentiment analysis
• Personalized product recommendations based on browsing history
• Live chat escalation system for complex issues
• Customer journey visualization and optimization suggestions
Tips: Try AI-powered web application platform such as lovable

E5: EDUCATIONAL CONTENT INTELLIGENCE PLATFORM

Problem Description:
University with 15,000 students struggles with course content management and student learning outcomes. Faculty spend 20+ hours weekly creating study materials from textbooks and research papers. Students have difficulty connecting concepts across courses, leading to 30% course failure rates and poor knowledge retention.

Expected Functionalities:
• Upload textbooks and research papers to generate study guides and summaries
• Create personalized learning materials based on student performance data
• Generate quiz questions and practice tests from course content
• Cross-reference concepts between different subjects and courses
• Extract key learning objectives and outcomes from academic documents
• Create interactive study notes with concept mapping
• Generate course syllabi recommendations based on academic standards
• Provide AI-powered tutoring responses based on uploaded course materials
Tips: Try AI-powered tool such as NotebookLM

1. AI for Industry

1. AI-Powered Retail Inventory Optimization

Summary: Develop an AI system that predicts optimal inventory levels for retail stores by analyzing sales patterns, seasonal trends, and external factors to minimize stockouts and overstock situations.

Problem Statement: Retail businesses struggle with inventory management, leading to lost sales from stockouts or increased costs from overstocking. Your task is to build an AI solution that analyzes historical sales data, weather patterns, local events, and market trends to predict optimal inventory levels for different product categories. The system should provide real-time recommendations for reordering, identify slow-moving items, and suggest promotional strategies to optimize inventory turnover while maintaining customer satisfaction.

Steps:

- Design a data ingestion pipeline that accepts multiple data sources including POS systems, weather APIs, and event calendars
- Implement machine learning models (ARIMA, LSTM, or Prophet) to forecast demand for different product categories
- Create a recommendation engine that suggests optimal stock levels, reorder points, and safety stock quantities
- Build a dashboard interface showing current inventory status, predicted demand, and actionable recommendations
- Develop alert systems for critical inventory situations and automated reordering capabilities
- Include scenario analysis tools to evaluate impact of promotions, seasonality, and external events

Suggested Data Requirements:
- Historical sales data with timestamps, product IDs, quantities, and prices (12+ months)
- Product catalog with categories, seasonality flags, and supplier information
- Weather data and local event calendars
- Store location and demographic data

Themes: AI for Industry, Classical AI/ML/DL for prediction

The steps and data requirements outlined above are intended solely as reference points to assist you in conceptualising your solution.

2. Manufacturing Quality Control AI Vision System

Summary: Create an AI-powered computer vision system that automatically detects defects in manufactured products on assembly lines, reducing quality control costs and improving product consistency.

Problem Statement: Manufacturing companies face challenges in maintaining consistent product quality while managing inspection costs and speed. Your task is to develop a computer vision AI system that can identify various types of defects (scratches, dents, color variations, dimensional issues) in real-time during production. The system should integrate with existing manufacturing equipment, provide instant feedback to operators, and maintain detailed quality metrics for continuous improvement.

Steps:
- Design a computer vision pipeline using CNNs or Vision Transformers for defect detection
- Implement real-time image processing capabilities for assembly line integration
- Create a classification system for different defect types with confidence scoring
- Build an operator interface showing real-time quality status and defect locations
- Develop a feedback loop system for continuous model improvement with human validation
- Include statistical quality control charts and trend analysis capabilities

Suggested Data Requirements:
- High-resolution images of products (both defective and non-defective samples, 1000+ each category)
- Defect classification labels and severity ratings

- Production line metadata (timestamps, batch numbers, operator IDs)

Themes: AI for Industry, Classical AI/ML/DL for prediction

The steps and data requirements outlined above are intended solely as reference points to assist you in conceptualising your solution.

3. Banking Fraud Detection Real-Time Analytics

Summary: Develop a real-time fraud detection system for banking transactions that combines machine learning algorithms with rule-based engines to identify suspicious activities and minimize false positives.

Problem Statement: Financial institutions need to detect fraudulent transactions in real-time while minimizing disruption to legitimate customers. Your task is to build a comprehensive fraud detection system that analyzes transaction patterns, user behavior, and contextual information to identify potentially fraudulent activities. The system should provide risk scores, explanations for decisions, and adaptive learning capabilities to evolve with new fraud patterns.

Steps:

- Design a real-time data processing pipeline for transaction analysis
- Implement multiple ML models (Random Forest, Isolation Forest, Neural Networks) for fraud detection
- Create a rule engine for known fraud patterns and regulatory compliance
- Build a risk scoring system with explainable AI features
- Develop an alert management system for fraud analysts with case prioritization
- Include model monitoring and adaptive learning capabilities for new fraud patterns

Suggested Data Requirements:

- Historical transaction data with fraud labels (100K+ transactions)
- User profile information and behavioral patterns
- Merchant and location data
- Device and session information for online transactions

Themes: AI for Industry, Classical AI/ML/DL for prediction

The steps and data requirements outlined above are intended solely as reference points to assist you in conceptualising your solution.

2. AI in Service Lines

4. IoT Predictive Maintenance Platform

Summary: Develop an AI platform that analyzes IoT sensor data from industrial equipment to predict failures, schedule maintenance, and optimize equipment performance.

Problem Statement: Industrial IoT deployments generate vast amounts of sensor data that can predict equipment failures before they occur. Your task is to build a predictive maintenance platform that processes real-time sensor data, identifies patterns indicating potential failures, and recommends optimal maintenance schedules. The system should support multiple equipment types, provide cost-benefit analysis for maintenance decisions, and integrate with existing maintenance management systems.

Steps:

- Design a scalable data ingestion pipeline for IoT sensor streams.
- Implement anomaly detection algorithms for various sensor types (vibration, temperature, pressure)
- Create predictive models for different failure modes using time series analysis
- Build a maintenance scheduling optimization engine considering costs and priorities
- Develop a monitoring dashboard with real-time equipment health status
- Include mobile alerts and work order integration for maintenance teams

Suggested Data Requirements:

- Time-series sensor data from multiple equipment types (temperature, vibration, pressure, etc.). You could generate synthetic data
- Equipment maintenance history and failure records
- Spare parts inventory and cost information
- Equipment specifications and operating parameters

Themes: AI in Service lines, Classical AI/ML/DL for prediction

The steps and data requirements outlined above are intended solely as reference points to assist you in conceptualising your solution.

3. AI for TCS

5. HR Talent Matching and Development AI

Summary: Build an AI system that matches employees with optimal projects and career opportunities while identifying skill gaps and recommending personalized development paths.

Problem Statement: Organizations need to efficiently match talent with projects while supporting employee growth and retention. Your task is to create an AI platform that analyzes employee skills, preferences, and career aspirations to recommend optimal project assignments and development opportunities. The system should consider project requirements, team dynamics, and organizational goals while providing personalized career guidance and skill development recommendations.

Steps:
- Design a skill assessment and profiling system using NLP for resume and project analysis
- Implement matching algorithms considering skills, preferences, and growth opportunities
- Create a recommendation engine for training and development programs
- Build career path visualization and planning tools
- Develop project staffing optimization with constraint satisfaction
- Include performance prediction and success probability scoring

Suggested Data Requirements:
- Employee profiles with skills, experience, and preferences
- Project requirements and historical staffing data
- Training course catalogs and completion records
- Performance review data and career progression histories

Themes: AI for TCS, Classical AI/ML/DL for prediction

The steps and data requirements outlined above are intended solely as reference points to assist you in conceptualising your solution.

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6. Finance Spend Analytics and Budget Optimization

Summary: Develop an AI solution that analyzes organizational spending patterns, identifies cost optimization opportunities, and provides intelligent budget planning recommendations.

Problem Statement: Financial departments need better insights into spending patterns and more accurate budget forecasting. Your task is to build an AI system that analyzes historical spending data, identifies anomalies and optimization opportunities, and provides predictive budget recommendations. The system should categorize expenses, benchmark against industry standards, and suggest cost reduction strategies while maintaining operational effectiveness.

Steps:

- Design automated expense categorization using ML classification
- Implement anomaly detection for unusual spending patterns
- Create forecasting models for budget planning and variance analysis
- Build benchmarking capabilities against industry and historical data
- Develop cost optimization recommendations with impact analysis
- Include approval workflow integration and spend compliance monitoring

Suggested Data Requirements:

- Historical financial transaction data with categories and departments
- Budget vs actual spending comparisons
- Vendor and contract information
- Industry benchmarking data and economic indicators

Themes: AI for TCS, Classical AI/ML/DL for prediction

The steps and data requirements outlined above are intended solely as reference points to assist you in conceptualising your solution.

7. Sales Performance Analytics and Lead Scoring

Summary: Create an AI platform that analyzes sales performance, scores leads based on conversion probability, and provides personalized recommendations for sales teams.

Problem Statement: Sales organizations need better lead qualification and performance insights to maximize revenue. Your task is to develop an AI system that scores leads based on conversion likelihood, analyzes sales performance patterns, and provides actionable recommendations for sales teams. The system should integrate with CRM systems, predict deal closure probability, and recommend optimal sales strategies for different customer segments.

Steps:
- Design lead scoring algorithms using customer data and behavioral signals
- Implement sales forecasting models for pipeline management
- Create performance analytics dashboards for individual and team metrics
- Build recommendation engines for optimal sales strategies and timing
- Develop customer segmentation and targeting capabilities
- Include A/B testing framework for sales strategy optimization

Suggested Data Requirements:
- CRM data with leads, opportunities, and customer interactions
- Sales performance metrics and quota achievement records
- Customer demographic and firmographic data
- Email and communication logs with engagement metrics-

Themes: AI for TCS, Classical AI/ML/DL for prediction

The steps and data requirements outlined above are intended solely as reference points to assist you in conceptualising your solution.

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4. Virtual Workers / Copilots

8. Intelligent Code Review Copilot

Summary: Develop an AI copilot that assists developers with code reviews by identifying potential issues, suggesting improvements, and ensuring coding standards compliance.

Problem Statement: Code reviews are time-consuming and often miss subtle issues or inconsistencies. Your task is to create an AI copilot that automatically reviews code changes, identifies potential bugs, security vulnerabilities, and performance issues while suggesting improvements. The system should learn from existing code patterns, enforce coding standards, and provide educational feedback to help developers improve their skills.
Steps:
- Design a code analysis engine using AST parsing and pattern recognition
- Implement ML models trained on code quality metrics and bug patterns
- Create integration with version control systems (Git, GitHub, GitLab)

- Build an intelligent feedback system with severity classification
- Develop learning capabilities from human reviewer feedback
- Include coding standard enforcement and best practice recommendations

Suggested Data Requirements:
- Code repositories with commit history and review comments
- Bug tracking data linked to code changes
- Coding standards and style guide documentation
- Security vulnerability databases and patterns

Themes: Virtual workers / Copilots, AI in software engineering lifecycle

The steps and data requirements outlined above are intended solely as reference points to assist you in conceptualising your solution.

9. Document Intelligence Copilot

Summary: Build an AI copilot that helps users create, analyze, and manage documents by providing intelligent suggestions, content extraction, and automated formatting.

Problem Statement: Knowledge workers spend significant time on document creation and management tasks. Your task is to develop a document intelligence copilot that assists with content creation, formatting, fact-checking, and information extraction from various document types. The system should understand document context, suggest relevant content, automate formatting tasks, and help maintain consistency across organizational documents.
Steps:
- Design document parsing capabilities for multiple formats (PDF, Word, PowerPoint)
- Implement NLP/LLM models for content understanding and generation
- Create intelligent formatting and style consistency tools
- Build fact-checking and citation verification capabilities
- Develop template suggestion and content recommendation features
- Include collaboration tools with version control and change tracking

Suggested Data Requirements:
- Sample documents in various formats and types
- Organizational templates and style guides
- Reference databases for fact-checking

- User interaction logs and feedback data

Themes: Virtual workers / Copilots, GenAI & its techniques

The steps and data requirements outlined above are intended solely as reference points to assist you in conceptualising your solution.

10. Meeting Intelligence Assistant

Summary: Create an AI assistant that enhances meeting productivity by providing meeting transcription, action item extraction, and follow-up automation.

Problem Statement: Meetings often lack proper documentation and follow-through on action items. Your task is to build an AI meeting assistant that provides transcription, identifies key decisions and action items, and automates follow-up tasks. The system should generate intelligent meeting summaries, and track action item completion across the organization.

Steps:

- Design text (transcript) processing with speaker identification
- Implement NLP models for action item and decision extraction
- Build automated summary generation with key highlights
- Develop follow-up tracking and reminder systems
- Include meeting analytics and productivity metrics

Suggested Data Requirements:

- Transcripts of Audio recordings of the meetings
- Meeting agenda templates and structures
- Action item tracking data and completion rates
- Calendar integration data and attendee information

Themes: Virtual workers / Copilots, Multi-modal UX

The steps and data requirements outlined above are intended solely as reference points to assist you in conceptualising your solution.

5. Edge AI

11. Smart Retail Edge Vision System

Summary: Create an AI system for retail environments that analyzes customer behavior, manages inventory, and provides personalized shopping experiences using in-store cameras.

Problem Statement: Retail stores need real-time customer analytics and inventory management without privacy concerns of cloud-based processing. Your task is to develop an AI solution that analyzes customer traffic patterns, monitors inventory levels, and provides personalized recommendations. The system should respect privacy requirements while delivering actionable business insights.

Steps:
- Design privacy-preserving computer vision algorithms for customer analytics
- Implement real-time inventory tracking using shelf-mounted cameras
- Create customer traffic flow analysis and heat mapping
- Build recommendation engine based on customer behavior patterns
- Develop integration with POS systems and inventory management
- Include store optimization recommendations and A/B testing capabilities

Suggested Data Requirements:
- Retail store images with privacy considerations
- Product catalog and inventory data
- Customer transaction histories (anonymized)
- Store layout and fixture information

Themes: Edge AI, AI for Industry

The steps and data requirements outlined above are intended solely as reference points to assist you in conceptualising your solution.

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6. Agents & APIs

12. Multi-Platform Social Media Management Agent

Summary: Develop an AI agent that manages social media presence across multiple platforms by creating content, scheduling posts, and engaging with audiences based on brand guidelines.

Problem Statement: Managing social media presence across multiple platforms is time-consuming and requires consistent brand voice. Your task is to create an AI agent that generates platform-specific content, schedules posts optimally, responds to comments and messages, and provides analytics insights. The system should maintain brand consistency while adapting to each platform's unique requirements and audience preferences.

Steps:

- Design content generation engines for different social media platforms
- Implement scheduling optimization based on audience engagement patterns
- Create automated response systems for common customer inquiries
- Build brand voice consistency checking and content approval workflows
- Develop cross-platform analytics and performance tracking
- Include crisis management and escalation protocols for sensitive situations

Suggested Data Requirements:

- Social media post history and engagement metrics
- Brand guidelines and approved content templates
- Customer interaction logs and response examples
- Competitor analysis data and industry benchmarks

Themes: Agents & APIs, GenAI & its techniques

The steps and data requirements outlined above are intended solely as reference points to assist you in conceptualising your solution.

13. E-commerce Customer Service Agent

Summary: Build an intelligent customer service agent that handles e-commerce inquiries, processes returns, tracks orders, and escalates complex issues to human agents.

Problem Statement: E-commerce businesses need to provide 24/7 customer support while managing high volumes of routine inquiries. Your task is to develop an AI agent that handles common customer service tasks including order tracking, return processing, product recommendations, and issue resolution.

Steps:

- Design conversational AI with intent recognition for e-commerce scenarios
- Build order tracking and delivery status communication

- Develop escalation logic for complex issues requiring human intervention
- Include customer satisfaction monitoring and feedback collection

Suggested Data Requirements:

- Customer service chat logs and email interactions
- E-commerce transaction and order data
- Product catalogs with detailed specifications
- Return and refund policy documentation

Themes: Agents & APIs, Virtual workers / Copilots

The steps and data requirements outlined above are intended solely as reference points to assist you in conceptualising your solution.

14. Financial Advisory AI Agent

Summary: Create an AI agent that provides personalized financial advice, portfolio recommendations, and investment insights based on individual risk profiles and market conditions.

Problem Statement: Individual investors need personalized financial guidance but lack access to professional advisory services. Your task is to build an AI financial advisor that analyzes personal financial data, assesses risk tolerance, and provides investment recommendations. The system should consider market conditions, regulatory requirements, and individual goals while providing educational content and continuous portfolio monitoring.

Steps:

- Design risk assessment questionnaires and profile analysis
- Implement portfolio optimization algorithms with diversification strategies
- Create market analysis and trend identification capabilities
- Build regulatory compliance checking for investment recommendations
- Develop educational content delivery based on user knowledge level
- Include performance tracking and rebalancing recommendations

Suggested Data Requirements:

- Personal financial data templates and risk assessment profiles
- Historical market data and investment performance metrics
- Regulatory guidelines and compliance requirements

- Educational content and investment strategy documentation

Themes: Agents & APIs, AI for Industry

The steps and data requirements outlined above are intended solely as reference points to assist you in conceptualising your solution.

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8. Classical AI/ML/DL for prediction

15. Healthcare Patient Risk Stratification System

Summary: Develop a machine learning system that analyzes patient data to predict health risks and recommend preventive care interventions for healthcare providers.

Problem Statement: Healthcare providers need to identify high-risk patients for proactive care management. Your task is to build a predictive model that analyzes patient demographics, medical history, lab results, and lifestyle factors to calculate risk scores for various health conditions. The system should provide actionable recommendations for preventive care while ensuring regulatory compliance and clinical interpretability.

Steps:

- Design feature engineering pipeline for multi-source healthcare data
- Implement ensemble models combining different ML algorithms for risk prediction
- Create interpretable model outputs with SHAP values and feature importance
- Build clinical decision support interface with risk stratification
- Develop care plan recommendation engine based on evidence-based guidelines
- Include model validation and regulatory compliance documentation

Suggested Data Requirements:

- De-identified patient records with demographics and medical history
- Lab results and vital signs data
- Treatment outcomes and readmission records
- Clinical guidelines and care protocol documentation

Themes: Classical AI/ML/DL for prediction, AI for Industry

The steps and data requirements outlined above are intended solely as reference points to assist you in conceptualising your solution.

16. Supply Chain Demand Forecasting Platform

Summary: Build a comprehensive demand forecasting system that predicts product demand across multiple locations and time horizons using various machine learning techniques.

Problem Statement: Supply chain optimization requires accurate demand forecasting across different products, locations, and time periods. Your task is to create a forecasting platform that combines multiple algorithms to predict demand while considering seasonality, promotions, external factors, and supply constraints. The system should provide confidence intervals, scenario analysis, and integration with inventory management systems.

Steps:

- Design time series preprocessing pipeline with feature engineering
- Implement multiple forecasting models (ARIMA, Prophet, LSTM, XGBoost)
- Create model ensemble and selection logic based on data characteristics
- Build scenario analysis capabilities for promotions and external events
- Develop forecast accuracy monitoring and model retraining automation
- Include integration with inventory optimization and procurement systems

Suggested Data Requirements:

- Historical sales data with product, location, and time dimensions
- Promotional calendar and marketing campaign data
- External factors (weather, economic indicators, events)
- Supply chain constraints and lead time information

Themes: Classical AI/ML/DL for prediction, AI in Service lines

The steps and data requirements outlined above are intended solely as reference points to assist you in conceptualising your solution.

17. Content Recommendation Engine

Summary: Create a personalized content recommendation system that combines collaborative filtering, content-based filtering, and deep learning to suggest relevant articles, videos, or products.

Problem Statement: Content platforms need to provide personalized recommendations to increase user engagement and satisfaction. Your task is to build a hybrid recommendation system that analyzes user behavior, content features, and contextual information to suggest relevant items. The system should handle cold-start problems, provide diverse recommendations, and adapt to changing user preferences in real-time.

Steps:

- Design user and item feature extraction pipelines
- Implement multiple recommendation algorithms (collaborative filtering, content-based, matrix factorization)
- Create deep learning models for complex pattern recognition in user behavior
- Build real-time inference system with A/B testing capabilities
- Develop recommendation explanation and diversity optimization
- Include performance monitoring and recommendation quality metrics

Suggested Data Requirements:

- User interaction data (clicks, views, ratings, time spent)
- Content metadata (categories, tags, descriptions, features)
- User demographic and preference profiles
- Contextual information (time, device, location)

Themes: Classical AI/ML/DL for prediction, AI for Industry

The steps and data requirements outlined above are intended solely as reference points to assist you in conceptualising your solution.

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9. GenAI & its techniques

18. RAG-Based Technical Documentation Assistant

Summary: Develop a RAG (Retrieval-Augmented Generation) system that helps users find information and generate responses from large technical documentation repositories.

Problem Statement: Technical documentation is often extensive and difficult to navigate, leading to inefficient information retrieval. Your task is to build a RAG system that indexes technical documents, retrieves relevant sections based on user queries, and generates accurate, contextual responses. The system should handle various document formats, maintain source attribution, and provide up-to-date information from evolving documentation.

Steps:
- Design document ingestion pipeline supporting multiple formats (PDF, HTML, Markdown)
- Implement advanced text chunking and embedding strategies for optimal retrieval
- Create semantic search capabilities using vector databases (FAISS, Pinecone)
- Build context-aware response generation with source attribution
- Develop query understanding and intent classification for technical domains
- Include document versioning and update mechanisms for evolving content
- Create evaluation metrics for retrieval quality and response accuracy

Suggested Data Requirements:
- Technical documentation in various formats (API docs, user manuals, code documentation)
- User query logs and information-seeking patterns
- Expert-validated question-answer pairs for evaluation
- Document metadata and versioning information

Themes: GenAI & its techniques, RAG

The steps and data requirements outlined above are intended solely as reference points to assist you in conceptualising your solution.

19. Prompt Engineering Optimization Platform

Summary: Build a platform that automatically optimizes prompts for language models by testing variations, analyzing performance, and generating improved prompt templates.

Problem Statement: Effective prompt engineering requires extensive experimentation and domain expertise. Your task is to create a platform that automates prompt optimization by generating variations, testing them against evaluation criteria, and iteratively improving prompt performance. The system should support different LLM models, provide performance analytics, and maintain a library of optimized prompts for various use cases.

Steps:
- Design prompt variation generation using genetic algorithms and template systems
- Implement automated testing framework with multiple evaluation metrics
- Create performance analytics dashboard comparing prompt effectiveness

- Build prompt template library with categorization and search capabilities
- Develop A/B testing infrastructure for prompt comparison
- Include integration with multiple LLM providers and cost optimization

Suggested Data Requirements:

- Prompt templates and examples across different domains
- Evaluation datasets with ground truth responses
- Performance metrics and user feedback data
- LLM API response logs and cost information

Themes: GenAI & its techniques, Prompt engineering & Meta prompts

The steps and data requirements outlined above are intended solely as reference points to assist you in conceptualising your solution.

20. Knowledge Graph Enhanced Q&A System

Summary: Create a question-answering system that combines knowledge graphs with generative AI to provide accurate, structured responses with reasoning chains.

Problem Statement: Traditional Q&A systems often lack structured reasoning and relationship understanding. Your task is to build a system that combines knowledge graphs with generative AI to answer complex questions requiring multi-hop reasoning. The system should construct and query knowledge graphs, generate explanations for answers, and provide confidence scores based on knowledge graph completeness.

Steps:

- Design knowledge graph construction from unstructured text using NER and relation extraction
- Implement graph-based query processing for multi-hop reasoning
- Create integration between graph queries and generative AI responses
- Build explanation generation showing reasoning paths through the knowledge graph
- Develop confidence scoring based on graph connectivity and source reliability
- Include graph visualization and interactive exploration capabilities

Suggested Data Requirements:

- Structured and unstructured text data for knowledge extraction
- Curated question-answer pairs requiring multi-hop reasoning

- Entity and relationship ontologies for domain-specific knowledge
- Source credibility and reliability metadata

Themes: GenAI & its techniques, Knowledge Graph, Graph RAG

The steps and data requirements outlined above are intended solely as reference points to assist you in conceptualising your solution.

21. Model Quantization and Fine-tuning Platform

Summary: Develop a platform that enables efficient model quantization and fine-tuning for deploying large language models on resource-constrained environments.

Problem Statement: Large language models require significant computational resources, limiting their deployment in edge environments. Your task is to create a platform that automates model quantization, fine-tuning, and optimization for specific use cases while maintaining performance quality. The system should support various quantization techniques, provide performance benchmarking, and enable easy deployment to different hardware configurations.

Steps:

- Design automated quantization pipeline supporting multiple techniques (INT8, INT4, dynamic)
- Implement fine-tuning workflows with parameter-efficient methods (LoRA, QLoRA)
- Create performance benchmarking suite measuring accuracy, speed, and memory usage
- Build deployment optimization for different hardware targets (CPU, GPU, mobile)
- Develop model comparison and selection tools based on constraints
- Include monitoring and quality assessment for quantized models

Suggested Data Requirements:

- Pre-trained model checkpoints and configuration files
- Domain-specific fine-tuning datasets
- Hardware performance benchmarks and constraints
- Quality evaluation datasets for model comparison

Themes: GenAI & its techniques, Quantization, Fine-tuning

The steps and data requirements outlined above are intended solely as reference points to assist you in conceptualising your solution.

8. Agentic AI

22. Multi-Step Research Assistant Agent

Summary: Build an autonomous research agent that can plan, execute, and synthesize multi-step research tasks by coordinating web searches, document analysis, and report generation.

Problem Statement: Complex research tasks require coordination of multiple information sources and analytical steps. Your task is to create an agentic AI system that can break down research queries into sub-tasks, execute searches across multiple sources, analyze findings, and synthesize comprehensive reports. The system should demonstrate reasoning, planning, and adaptive execution while maintaining source credibility and factual accuracy.

Steps:

- Design task decomposition and planning algorithms for research workflows
- Implement multi-source information retrieval (web search, database queries, document analysis)
- Create reasoning chains for connecting disparate information pieces
- Build synthesis and report generation capabilities with source attribution
- Develop fact-checking and credibility assessment mechanisms
- Include interactive refinement and follow-up question capabilities

Suggested Data Requirements:

- Sample research questions across various domains
- Diverse information sources (web pages, academic papers, databases)
- Fact-checking datasets and credibility scoring criteria
- Report templates and evaluation criteria

Themes: Agentic AI, Agents & APIs

The steps and data requirements outlined above are intended solely as reference points to assist you in conceptualising your solution.

23. Autonomous Data Analysis Agent

Summary: Create an AI agent that automatically performs exploratory data analysis, identifies patterns, generates insights, and creates visualizations without human intervention.

Problem Statement: Data analysis often requires significant manual effort and domain expertise. Your task is to build an autonomous agent that can explore datasets, identify relevant patterns, perform statistical analysis, and generate actionable insights. The system should adapt its analysis approach based on data characteristics, create appropriate visualizations, and provide narrative explanations of findings.

Steps:

- Design automated data profiling and quality assessment capabilities
- Implement adaptive analysis strategy selection based on data types and patterns
- Create automated statistical testing and hypothesis generation
- Build dynamic visualization generation with narrative explanations
- Develop insight ranking and significance assessment algorithms
- Include interactive drill-down and follow-up analysis capabilities

Suggested Data Requirements:

- Diverse datasets with different structures and domains
- Analysis templates and best practices documentation
- Statistical test catalogs and interpretation guidelines
- Visualization examples and design patterns

Themes: Agentic AI, Classical AI/ML/DL for prediction

The steps and data requirements outlined above are intended solely as reference points to assist you in conceptualising your solution.

24. Business Process Automation Agent

Summary: Develop an agentic AI system that can understand business processes, identify automation opportunities, and implement workflow solutions autonomously.

Problem Statement: Business process automation requires deep understanding of workflows and decision-making logic. Your task is to create an AI agent that can analyze business processes, identify inefficiencies, design automation solutions, and implement workflow improvements. The system should understand process documentation, interact with various business systems, and continuously optimize based on performance feedback.

Steps:

- Design process mining capabilities to extract workflows from operational data

- Implement decision logic understanding and automation rule generation
- Create integration framework for various business applications and APIs
- Build workflow optimization algorithms considering constraints and objectives
- Develop monitoring and performance feedback loops for continuous improvement
- Include change management and rollback capabilities for automation updates

Suggested Data Requirements:

- Business process documentation and workflow diagrams
- Operational data logs showing process execution patterns
- System integration schemas and API documentation
- Performance metrics and optimization criteria

Themes: Agentic AI, AI in Service lines

The steps and data requirements outlined above are intended solely as reference points to assist you in conceptualising your solution.

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9. Orchestration & MCP

25. Multi-Agent Coordination Platform

Summary: Build a platform that orchestrates multiple AI agents to collaborate on complex tasks requiring diverse skills and knowledge domains.

Problem Statement: Complex problems often require multiple specialized capabilities that exceed single agent limitations. Your task is to create an orchestration platform that coordinates multiple AI agents, each with specialized skills, to collaboratively solve complex problems. The system should handle task delegation, inter-agent communication, conflict resolution, and result synthesis while maintaining efficient resource utilization.

Steps:

- Design agent registry and capability discovery mechanisms
- Implement task decomposition and delegation algorithms
- Create inter-agent communication protocols and message passing
- Build conflict resolution and consensus-building capabilities
- Develop resource scheduling and load balancing for agent coordination

- Include monitoring and debugging tools for multi-agent interactions

Suggested Data Requirements:

- Agent capability profiles and performance metrics
- Complex task examples requiring multi-agent collaboration
- Communication protocol specifications and message templates
- Resource utilization and performance benchmarks

Themes: Orchestration & MCP, Agentic AI

The steps and data requirements outlined above are intended solely as reference points to assist you in conceptualising your solution.

26. Intelligent Task Routing System

Summary: Develop a system that intelligently routes tasks to appropriate AI agents or human workers based on complexity, expertise requirements, and current workload.

Problem Statement: Optimal task assignment requires understanding task requirements and available resources. Your task is to create an intelligent routing system that analyzes incoming tasks, determines optimal assignment strategies, and dynamically routes work to AI agents or human workers. The system should consider skill matching, workload balancing, quality requirements, and cost optimization while maintaining service level agreements.

Steps:

- Design task analysis and classification system for routing decisions
- Implement skill matching algorithms for optimal resource assignment
- Create dynamic load balancing and queue management capabilities
- Build quality prediction and SLA monitoring systems
- Develop cost optimization algorithms considering resource costs and performance
- Include escalation and reassignment mechanisms for complex scenarios

Suggested Data Requirements:

- Task descriptions with complexity and skill requirement annotations
- Resource profiles including AI agent capabilities and human expertise
- Historical performance data and quality metrics
- Cost information and SLA requirements

Themes: Orchestration & MCP, AI in Service lines

The steps and data requirements outlined above are intended solely as reference points to assist you in conceptualising your solution.

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10. Deep-tech Research

27. Mixture of Experts Model Implementation

Summary: Build a Mixture of Experts (MoE) model architecture that dynamically routes inputs to specialized expert networks for improved efficiency and performance.

Problem Statement: Large models consume significant computational resources even for simple tasks. Your task is to implement a Mixture of Experts architecture that activates only relevant expert networks based on input characteristics, improving computational efficiency while maintaining or enhancing model performance. The system should include expert specialization, routing mechanisms, and load balancing across experts.

Steps:
- Design expert network architectures with domain specialization
- Implement gating mechanisms for intelligent expert selection
- Create training procedures for expert specialization and load balancing
- Build inference optimization for efficient expert activation
- Develop monitoring tools for expert utilization and performance analysis
- Include techniques for expert pruning and dynamic scaling

Suggested Data Requirements:
- Large-scale training datasets with diverse task types
- Domain-specific datasets for expert specialization
- Performance benchmarks for efficiency and quality comparison
- Expert utilization patterns and load distribution data

Themes: Deep-tech research, Mixture of experts

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28. Explainable AI Dashboard for Complex Models

Summary: Develop a comprehensive explainability platform that provides interpretable insights into complex AI model decisions across different model types and domains.

Problem Statement: Complex AI models operate as black boxes, limiting trust and regulatory compliance. Your task is to create a unified explainability platform that works across different model types (neural networks, ensembles, LLMs) and provides various explanation methods (SHAP, LIME, attention visualization, counterfactuals). The system should offer both technical and non-technical explanations while ensuring explanation quality and consistency.
Steps:

- Design modular explanation framework supporting multiple methods and model types
- Implement attention visualization and feature importance analysis
- Create counterfactual explanation generation and what-if analysis tools
- Build user-friendly interfaces for different stakeholder needs (technical, business, regulatory)
- Develop explanation quality metrics and validation frameworks
- Include explanation aggregation and comparison capabilities across different methods

Suggested Data Requirements:

- Trained models across different architectures and domains
- Validation datasets with ground truth explanations
- User interaction data showing explanation preferences
- Regulatory compliance requirements and documentation

Themes: Deep-tech research, Explainability

The steps and data requirements outlined above are intended solely as reference points to assist you in conceptualising your solution.

29. Custom Domain-Specific Model Architecture

Summary: Build a framework for creating custom neural network architectures optimized for specific domain requirements and constraints.

Problem Statement: Standard model architectures may not be optimal for specialized domains with unique requirements. Your task is to create a framework that enables rapid prototyping and optimization of custom neural network architectures for specific domains. The system should support architecture search, performance optimization, and automated hyperparameter tuning while considering domain-specific constraints and objectives.
Steps:

- Design neural architecture search (NAS) algorithms for domain-specific optimization

- Implement modular building blocks for rapid architecture prototyping
- Create domain-specific loss functions and evaluation metrics
- Build automated hyperparameter optimization and training pipelines
- Develop performance profiling and bottleneck analysis tools
- Include transfer learning and knowledge distillation capabilities

Suggested Data Requirements:

- Domain-specific datasets with performance requirements
- Architecture performance benchmarks and constraints
- Hardware specifications and deployment targets
- Domain expert knowledge and constraint definitions

Themes: Deep-tech research, Custom models for specialized purposes

The steps and data requirements outlined above are intended solely as reference points to assist you in conceptualising your solution.

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11. AI for Creative

30. AI-Powered Design Generation Platform

Summary: Create a platform that generates creative designs for various media (logos, posters, web layouts) based on user requirements and brand guidelines.

Problem Statement: Creative design requires balancing aesthetic appeal with functional requirements and brand consistency. Your task is to build an AI platform that generates designs across multiple formats, considers brand guidelines and user preferences, and provides iterative refinement capabilities. The system should understand design principles, maintain style consistency, and enable collaborative design workflows.

Steps:

- Design generative models for different design formats (logos, layouts, illustrations)
- Implement style transfer and brand guideline adherence mechanisms
- Create user preference learning and iterative refinement systems
- Build collaborative design tools with version control and feedback integration
- Develop design principle evaluation and quality assessment frameworks
- Include export capabilities for various formats and design software integration

Suggested Data Requirements:

- Design portfolio datasets across different styles and formats
- Brand guideline examples and style specifications
- User feedback and preference data
- Design principle documentation and best practices

Themes: AI for creative, GenAI & its techniques

The steps and data requirements outlined above are intended solely as reference points to assist you in conceptualising your solution.

31. 3D Model Generation and Customization Tool

Summary: Develop an AI system that generates and customizes 3D models based on text descriptions, sketches, or reference images for various applications.

Problem Statement: 3D model creation requires specialized skills and significant time investment. Your task is to create an AI tool that generates 3D models from natural language descriptions, 2D sketches, or reference images. The system should enable model customization, provide different levels of detail, and export models in various formats for different use cases (gaming, architecture, manufacturing).

Steps:

- Design text-to-3D and image-to-3D generation pipelines
- Implement model customization tools for geometry, textures, and materials
- Create level-of-detail generation for different application requirements
- Build export capabilities for various 3D formats (OBJ, FBX, GLTF)
- Develop quality assessment and optimization tools for generated models
- Include integration with popular 3D software and game engines

Suggested Data Requirements:

- 3D model datasets with text descriptions and metadata
- Reference image collections with corresponding 3D models
- Material and texture libraries
- Application-specific model requirements and constraints

Themes: AI for creative, 3D modelling

The steps and data requirements outlined above are intended solely as reference points to assist you in conceptualising your solution.

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32. Interactive Story Generation Platform

Summary: Build an AI platform that creates interactive, branching narratives with character development, plot consistency, and user choice integration.

Problem Statement: Interactive storytelling requires maintaining narrative coherence while adapting to user choices. Your task is to create a platform that generates engaging, branching stories with consistent character development, plot progression, and meaningful user interactions. The system should understand narrative structure, maintain story continuity across different paths, and provide rich character and world-building capabilities.

Steps:

- Design narrative generation models with plot structure understanding
- Implement character development tracking and consistency maintenance
- Create branching narrative logic with meaningful choice consequences
- Build world-building and setting consistency management
- Develop user engagement metrics and story quality assessment
- Include collaborative storytelling features and story sharing capabilities

Suggested Data Requirements:

- Interactive fiction and branching narrative examples
- Character development templates and relationship mapping
- Plot structure frameworks and story beats
- User choice data and engagement metrics

Themes: AI for creative, Content, Story-telling

The steps and data requirements outlined above are intended solely as reference points to assist you in conceptualising your solution.

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12. Open Source / Open Weight Models

33. Multi-Model Comparison and Benchmarking Platform

Summary: Develop a comprehensive platform for comparing and benchmarking open-source language models across various tasks and performance metrics.

Problem Statement: Selecting optimal open-source models requires systematic comparison across multiple dimensions. Your task is to create a benchmarking platform that evaluates open-source models across different tasks, measures performance metrics, and provides recommendations based on specific use case requirements. The system should automate model testing, provide fair comparisons, and maintain updated benchmarks as new models are released.

Steps:

- Design automated model loading and evaluation pipelines
- Implement comprehensive benchmark suites covering various tasks (reasoning, coding, creativity)
- Create performance metrics analysis including speed, accuracy, and resource usage
- Build recommendation engine for model selection based on requirements
- Develop cost-benefit analysis tools considering computational resources
- Include model fine-tuning comparison and specialized task evaluation

Suggested Data Requirements:

- Standardized benchmark datasets across different domains
- Model performance baselines and historical comparisons
- Hardware resource utilization data
- Task-specific evaluation criteria and scoring methods

Themes: Open source / Open weight models, Classical AI/ML/DL for prediction

The steps and data requirements outlined above are intended solely as reference points to assist you in conceptualising your solution.

34. Open Model Fine-tuning Pipeline

Summary: Build an automated pipeline for fine-tuning open-source language models on custom datasets with optimization for different hardware configurations.

Problem Statement: Fine-tuning large open-source models requires expertise and computational resources. Your task is to create an automated pipeline that simplifies fine-tuning of open-source models for specific tasks and domains. The system should handle data preparation, hyperparameter optimization, distributed training, and model evaluation while providing cost-effective solutions for different hardware setups.

Steps:

- Design automated data preprocessing and validation pipelines
- Implement parameter-efficient fine-tuning methods (LoRA, QLoRA, AdaLoRA)
- Create distributed training orchestration for multi-GPU setups
- Build hyperparameter optimization using Bayesian or evolutionary methods
- Develop model evaluation and comparison frameworks
- Include deployment optimization and model serving capabilities

Suggested Data Requirements:

- Domain-specific fine-tuning datasets
- Hardware configuration specifications and performance benchmarks
- Hyperparameter search spaces and optimization histories
- Model evaluation criteria and validation datasets

Themes: Open source / Open weight models, GenAI & its techniques

The steps and data requirements outlined above are intended solely as reference points to assist you in conceptualising your solution.

13. Proprietary Models

35. Advanced Prompt Template Management System

Summary: Create a comprehensive system for managing, versioning, and optimizing prompt templates across different proprietary LLM providers.

Problem Statement: Managing prompts across different providers and use cases requires systematic organization and optimization. Your task is to build a prompt management system that stores, versions, and optimizes prompt templates for various proprietary models. The system should enable A/B testing, performance tracking, and collaborative prompt development while maintaining security and compliance.

Steps:

- Design prompt template storage with versioning and metadata management
- Implement A/B testing framework for prompt optimization

- Create collaborative editing and review workflows for prompt development
- Build performance tracking and quality metrics across different providers
- Develop prompt security scanning and compliance checking
- Include automated prompt generation and improvement suggestions

Suggested Data Requirements:

- Prompt template libraries with performance metadata
- A/B testing results and optimization histories
- Security and compliance requirements for prompt content
- User collaboration and editing logs

Themes: Proprietary models, GenAI & its techniques

The steps and data requirements outlined above are intended solely as reference points to assist you in conceptualising your solution.

15. Responsible AI

36. AI Bias Detection and Mitigation Platform

Summary: Develop a comprehensive platform that detects, measures, and mitigates bias in AI models across different domains and protected characteristics.

Problem Statement: AI systems can perpetuate or amplify existing biases, leading to unfair outcomes. Your task is to create a platform that systematically detects bias in AI models, quantifies fairness metrics, and provides mitigation strategies. The system should work across different model types, support various fairness definitions, and provide actionable recommendations for bias reduction while maintaining model performance.

Steps:

- Design bias detection algorithms for different types of discrimination (statistical, individual, counterfactual)
- Implement fairness metrics calculation and monitoring across protected groups
- Create bias mitigation techniques including pre-processing, in-processing, and post-processing methods
- Build explanation tools showing sources and impacts of detected bias
- Develop continuous monitoring and alerting for bias drift in production models

- Include regulatory compliance checking and documentation generation

Suggested Data Requirements:

- Datasets with protected attribute labels for bias testing
- Fairness evaluation benchmarks and ground truth data
- Regulatory guidelines and compliance requirements
- Historical bias detection and mitigation case studies

Themes: Responsible AI, AI design that assures Security, Legal and Privacy requirements

The steps and data requirements outlined above are intended solely as reference points to assist you in conceptualising your solution.

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37. Privacy-Preserving AI Training Framework

Summary: Build a framework that enables AI model training while preserving data privacy through techniques like differential privacy, federated learning, and secure computation.

Problem Statement: Organizations need to train AI models on sensitive data while maintaining privacy and regulatory compliance. Your task is to create a framework that implements privacy-preserving techniques for AI training including differential privacy, federated learning, and secure multi-party computation. The system should provide privacy guarantees, enable collaborative learning without data sharing, and maintain model utility.

Steps:

- Design differential privacy mechanisms for various machine learning algorithms
- Create secure aggregation and communication protocols for multi-party computation
- Build privacy tracking systems
- Develop model utility assessment under different privacy constraints
- Include compliance reporting and privacy audit capabilities

Suggested Data Requirements:

- Sensitive training datasets with privacy requirements
- Model performance benchmarks under privacy constraints

Themes: Responsible AI, AI design that assures Security, Legal and Privacy requirements

The steps and data requirements outlined above are intended solely as reference points to assist you in conceptualising your solution.

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38. AI Governance and Compliance Monitoring System

Summary: Create a comprehensive governance system that monitors AI model deployment, ensures regulatory compliance, and manages AI risk across the organization.

Problem Statement: Organizations need systematic governance of AI systems to ensure compliance with regulations and ethical standards. Your task is to build a governance platform that monitors AI model deployment, tracks compliance with various regulations (GDPR, CCPA, AI Act), and provides risk management capabilities. The system should provide audit trails, policy enforcement, and automated compliance checking.

Steps:
- Design AI model registry and lifecycle tracking capabilities
- Implement automated compliance checking against multiple regulatory frameworks
- Create risk assessment and management workflows for AI deployments
- Build audit trail generation and documentation systems
- Develop policy enforcement mechanisms and approval workflows
- Include stakeholder notification and reporting dashboards

Suggested Data Requirements:
- AI model metadata and deployment information
- Regulatory framework requirements and compliance checklists
- Risk assessment criteria and scoring methodologies
- Audit requirements and documentation templates

Themes: Responsible AI, AI design that assures Security, Legal and Privacy requirements

The steps and data requirements outlined above are intended solely as reference points to assist you in conceptualising your solution.

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16. AI for Data & Data for AI

39. Synthetic Data Generation Platform

Summary: Develop a platform that generates high-quality synthetic data for AI training while preserving statistical properties and privacy characteristics of original datasets.

Problem Statement: Organizations need high-quality training data but face privacy, cost, and availability constraints. Your task is to create a synthetic data generation platform that produces realistic datasets maintaining statistical properties, relationships, and patterns of original data while ensuring privacy protection. The system should support various data types, provide quality assessment, and enable data augmentation for improved model training.

Steps:
- Design generative models for different data types (tabular, time-series, images, text)
- Implement statistical property preservation and relationship modeling
- Create privacy-preserving synthetic data generation with differential privacy
- Build quality assessment metrics and validation frameworks
- Develop data augmentation capabilities for improving model performance
- Include integration with popular ML training pipelines and data platforms

Suggested Data Requirements:
- Original datasets for synthesis reference and validation
- Statistical property specifications and relationship definitions
- Privacy requirements and synthetic data quality criteria
- ML model training requirements and performance benchmarks

Themes: AI for Data & Data for AI, Synthetic data generation

The steps and data requirements outlined above are intended solely as reference points to assist you in conceptualising your solution.

40. AI-Powered Data Quality and Cleaning Platform

Summary: Build an intelligent data quality platform that automatically detects, diagnoses, and corrects data quality issues using machine learning techniques.

Problem Statement: Data quality issues significantly impact AI model performance and business decisions. Your task is to create an AI-powered platform that automatically identifies data quality problems (missing values, outliers, inconsistencies, duplicates), diagnoses root causes, and provides intelligent correction suggestions. The system should learn from data patterns, adapt to domain-specific requirements, and provide transparent quality improvement processes.

Steps:
- Design automated data profiling and quality assessment algorithms

- Implement ML-based anomaly detection and outlier identification
- Create intelligent data imputation and correction mechanisms
- Build data lineage tracking and quality issue root cause analysis
- Develop domain-specific data validation rules and quality metrics
- Include data quality monitoring and alerting for production pipelines

Suggested Data Requirements:

- Datasets with known quality issues and correction examples
- Domain-specific data quality rules and validation criteria
- Historical data quality improvement cases and outcomes
- Data lineage information and processing pipeline metadata

Themes: AI for Data & Data for AI, Using AI for Data cleaning

The steps and data requirements outlined above are intended solely as reference points to assist you in conceptualising your solution.

41. Knowledge Graph Construction and Management Platform

Summary: Create a platform that automatically constructs and maintains knowledge graphs from unstructured data sources using NLP and graph algorithms.

Problem Statement: Organizations have vast amounts of unstructured data that contains valuable relationships and knowledge. Your task is to build a platform that automatically extracts entities, relationships, and concepts from text data to construct and maintain knowledge graphs. The system should handle multi-source data integration, resolve entity conflicts, and provide querying and visualization capabilities for knowledge discovery.

Steps:

- Design entity extraction and relationship identification using advanced NLP
- Implement entity resolution and knowledge graph fusion algorithms
- Create automated ontology generation and schema evolution capabilities
- Build graph querying interface with natural language support
- Develop knowledge graph visualization and exploration tools
- Include incremental updates and consistency maintenance mechanisms

Suggested Data Requirements:

- Unstructured text documents from various sources

- Existing knowledge bases and ontologies for reference
- Entity relationship examples and validation data
- Domain-specific vocabularies and concept hierarchies

Themes: AI for Data & Data for AI, Knowledge graphs

The steps and data requirements outlined above are intended solely as reference points to assist you in conceptualising your solution.

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17. AI for CyberSecurity & CyberSecurity for AI

42. AI-Powered Threat Detection and Response System

Summary: Develop an intelligent cybersecurity platform that uses machine learning to detect advanced threats, analyze attack patterns, and automate incident response.

Problem Statement: Modern cyber threats are sophisticated and evolve rapidly, requiring intelligent detection and response capabilities. Your task is to create an AI-powered cybersecurity platform that analyzes network traffic, system logs, and user behavior to detect threats in real-time. The system should identify both known and zero-day attacks, provide threat intelligence, and automate response actions while minimizing false positives.

Steps:

- Design multi-source data ingestion from network, endpoint, and cloud security tools
- Implement anomaly detection algorithms for identifying unusual behavior patterns
- Create threat classification and severity scoring using ensemble methods
- Build automated incident response workflows with human oversight
- Develop threat intelligence integration and attack pattern analysis
- Include forensic analysis tools and incident documentation automation

Suggested Data Requirements:

- Network traffic logs and security event data
- Known attack signatures and threat intelligence feeds
- User behavior baselines and normal activity patterns
- Incident response playbooks and escalation procedures

Themes: AI for CyberSecurity & CyberSecurity for AI, GenAI triaging for Cyberdefense

The steps and data requirements outlined above are intended solely as reference points to assist you in conceptualising your solution.

43. AI Model Security and Protection Platform

Summary: Build a comprehensive platform that protects AI models from adversarial attacks, data poisoning, and unauthorized access while ensuring model integrity.

Problem Statement: AI models face various security threats including adversarial attacks, model extraction, and data poisoning. Your task is to create a platform that implements security measures for AI models including adversarial defense, model watermarking, and access control. The system should detect attacks in real-time, provide model integrity verification, and ensure secure model deployment and serving.

Steps:

- Design adversarial attack detection and defense mechanisms
- Implement model watermarking and ownership verification systems
- Create secure model serving with access control and audit logging
- Build data lineage tracking and poisoning detection capabilities
- Develop model integrity monitoring and tampering detection
- Include secure model training environments with isolation and containerization

Suggested Data Requirements:

- Adversarial attack examples and defense validation data
- Model fingerprinting and watermarking validation datasets
- Access control policies and user authentication logs
- Security audit requirements and compliance standards

Themes: AI for CyberSecurity & CyberSecurity for AI, Training data confidentiality, Containerization & Isolation

The steps and data requirements outlined above are intended solely as reference points to assist you in conceptualising your solution.

44. AI-Based Malicious Code Detection

Problem Statement: Develop advanced AI models capable of detecting malicious code snippets, potential backdoors, and security threats in applications, including analysis of compiled or obfuscated code.

Steps:

1. **Code Pattern Analysis:** Identify malicious code patterns and signatures
2. **Backdoor Detection:** Discover hidden malicious functionality
3. **Obfuscation Analysis:** Analyze compiled and obfuscated code
4. **Threat Classification:** Categorize types of malicious code discovered
5. **Alert Generation:** Provide detailed findings and remediation guidance

Suggested Data Requirements:

- Malicious code samples and backdoor examples
- Obfuscated and compiled code analysis tools
- Code decompilation and reverse engineering frameworks
- Threat classification taxonomies
- Clean code baselines for comparison

The steps and data requirements outlined above are intended solely as reference points to assist you in conceptualising your solution.

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45. Automated Alert Prioritization and Triage

Problem Statement: Implement LLM models to intelligently categorize and prioritize security alerts from various sources (SIEM, EDR, network logs) based on severity, impact, and context to reduce alert fatigue and optimize human analyst efforts.

Steps:

1. **Alert Ingestion:** Collect alerts from multiple security sources
2. **Intelligent Classification:** Categorize alerts using AI-driven analysis
3. **Severity Assessment:** Evaluate true threat severity and impact
4. **Context Enrichment:** Add relevant contextual information
5. **Prioritization:** Rank alerts for analyst attention

Suggested Data Requirements:

- Multi-source security alert data (SIEM, EDR, network logs)
- Historical alert classification and outcome data
- Threat severity and impact assessment criteria
- Contextual information about assets and users
- Analyst feedback and triage decisions

The steps and data requirements outlined above are intended solely as reference points to assist you in conceptualising your solution.

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46. AI-Based Zero-Day Attack Detection

Problem Statement: Develop sophisticated Large Language Models capable of identifying previously unknown zero-day attacks by analyzing patterns, behaviors, and anomalies that traditional signature-based systems cannot detect.

Steps:

1. **Behavioral Analysis:** Monitor system behaviors for anomalies
2. **Pattern Recognition:** Identify novel attack patterns
3. **Zero-day Classification:** Distinguish zero-days from known threats
4. **Validation:** Confirm zero-day attack discoveries
5. **Response Coordination:** Alert and coordinate response efforts

Suggested Data Requirements:
- System behavior baselines and normal patterns
- Known attack signatures for comparison
- Anomaly detection training datasets
- Zero-day attack examples and characteristics
- Network and system log data

The steps and data requirements outlined above are intended solely as reference points to assist you in conceptualising your solution.

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47. Securing Agentic AI Frameworks

Problem Statement: Develop comprehensive security analysis and protection mechanisms for AI agents built using popular open-source frameworks such as AutoGen, LangGraph, CrewAI, CAMEL, and ChatDev.

Steps:

1. **Framework Analysis:** Study security implications of each framework
2. **Vulnerability Detection:** Identify framework-specific security issues
3. **Security Integration:** Embed security controls into agent workflows
4. **Testing:** Validate security measures across different frameworks
5. **Best Practices:** Develop security guidelines for framework users

Suggested Data Requirements:
- Framework documentation and source code

- Agent implementation examples
- Security testing results
- Vulnerability databases for each framework
- Developer security guidelines

The steps and data requirements outlined above are intended solely as reference points to assist you in conceptualising your solution.

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48. Securing Emerging Agentic AI Protocols

Problem Statement: Create a comprehensive security framework to analyze, monitor, and protect emerging AI agent communication protocols including MCP, A2A, and ACP from potential security threats and vulnerabilities.

Steps:

1. **Protocol Analysis:** Reverse engineer and understand protocol structures
2. **Vulnerability Detection:** Identify security weaknesses in communication patterns
3. **Threat Modeling:** Map potential attack vectors specific to agent protocols
4. **Security Implementation:** Develop protective measures and monitoring systems
5. **Validation:** Test security measures against simulated attacks

Suggested Data Requirements:

- Protocol specifications and documentation
- Network traffic samples from agent communications
- Known attack patterns on similar protocols
- Security testing frameworks and tools
- Agent behavior baselines and anomaly detection datasets

The steps and data requirements outlined above are intended solely as reference points to assist you in conceptualising your solution.

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18. AI Observability & FinOps for AI

49. Comprehensive AI Model Monitoring and Observability Platform

Summary: Create a comprehensive monitoring platform that tracks AI model performance, data drift, and operational metrics across the entire ML lifecycle.

Problem Statement: Production AI models require continuous monitoring to ensure performance, detect issues, and maintain reliability. Your task is to build an observability platform that monitors model predictions, data quality, performance metrics, and system health. The system should detect data drift, model degradation, and operational issues while providing actionable insights for model maintenance and improvement.

Steps:

- Design real-time model performance monitoring with customizable metrics
- Implement data drift detection using statistical and ML-based methods
- Create model explainability tracking and prediction analysis tools
- Build system performance monitoring for inference latency and resource usage
- Develop automated alerting and notification systems for various issue types
- Include model comparison and A/B testing capabilities for performance analysis

Suggested Data Requirements:

- Production model prediction logs and ground truth labels
- Historical performance baselines and drift detection thresholds
- System metrics and resource utilization data
- Model configuration and deployment metadata

Themes: AI Observability & FinOps for AI, Model performance, Activity & Accuracy insights, Anomaly detection

The steps and data requirements outlined above are intended solely as reference points to assist you in conceptualising your solution.

50. AI Cost Optimization and FinOps Platform

Summary: Develop a platform that tracks, analyzes, and optimizes costs associated with AI model development, training, and deployment across cloud and on-premise infrastructure.

Problem Statement: AI projects often have unpredictable and high infrastructure costs that need careful management and optimization. Your task is to create a FinOps platform that tracks costs across the AI lifecycle, identifies optimization opportunities, and provides recommendations for cost reduction. The system should monitor resource usage, predict costs, and enable budget management while maintaining performance requirements.

Steps:

- Design cost tracking integration with cloud providers and on-premise infrastructure
- Implement resource utilization analysis and optimization recommendations

- Create cost forecasting models based on usage patterns and project requirements
- Build budget management and alerting systems for cost control
- Develop cost allocation and chargeback mechanisms for different teams and projects
- Include ROI analysis and cost-benefit assessment tools for AI investments

Suggested Data Requirements:

- Cloud billing data and resource usage metrics
- AI workload patterns and resource requirements
- Performance benchmarks and cost-performance trade-offs
- Budget allocations and project cost targets

Themes: AI Observability & FinOps for AI, Cost visibility & optimization

The steps and data requirements outlined above are intended solely as reference points to assist you in conceptualising your solution.

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19. AI in Software Engineering Lifecycle

51. AI-Enhanced Code Generation and Review Platform

Summary: Build a comprehensive platform that assists developers with code generation, automated testing, and intelligent code review throughout the development lifecycle.

Problem Statement: Software development productivity can be significantly enhanced through AI assistance in coding tasks. Your task is to create a platform that generates code from natural language descriptions, automatically creates tests, and performs intelligent code reviews. The system should integrate with popular IDEs, understand project context, and provide suggestions that align with coding standards and best practices.

Steps:

- Design natural language to code generation with context awareness
- Implement automated test case generation and code coverage analysis
- Create intelligent code review with bug detection and optimization suggestions
- Build IDE integration plugins for seamless developer workflow
- Develop code documentation generation and maintenance tools
- Include code quality metrics and technical debt assessment

Suggested Data Requirements:

- Large code repositories with documentation and commit histories
- Test suites and code coverage information
- Code review comments and bug fix examples
- Coding standards and best practice documentation

Themes: AI in software engineering lifecycle, AI for Design-Build-Test-Deploy

The steps and data requirements outlined above are intended solely as reference points to assist you in conceptualising your solution.

52. Intelligent DevOps Automation Platform

Summary: Create an AI-powered DevOps platform that automates deployment pipelines, predicts failures, and optimizes infrastructure management.

Problem Statement: DevOps processes involve complex workflows that can benefit from intelligent automation and predictive insights. Your task is to build a platform that automates CI/CD pipelines, predicts deployment failures, and optimizes infrastructure resources. The system should learn from historical deployments, provide intelligent rollback decisions, and optimize resource allocation for cost and performance.

Steps:

- Design intelligent CI/CD pipeline automation with failure prediction
- Implement deployment risk assessment and automated rollback mechanisms
- Create infrastructure optimization algorithms for cost and performance
- Build monitoring integration for proactive issue detection and resolution
- Develop capacity planning and auto-scaling recommendations
- Include incident management automation and root cause analysis

Suggested Data Requirements:

- CI/CD pipeline execution logs and deployment histories
- Infrastructure metrics and resource utilization data
- Incident reports and resolution procedures
- Performance benchmarks and cost optimization targets

Themes: AI in software engineering lifecycle, AI in IT Ops

The steps and data requirements outlined above are intended solely as reference points to assist you in conceptualising your solution.

53. Legacy System Modernization Assistant

Summary: Develop an AI assistant that analyzes legacy codebases and provides recommendations for modernization, refactoring, and technology migration.

Problem Statement: Organizations struggle with modernizing legacy systems due to complexity and risk. Your task is to create an AI assistant that analyzes legacy code, identifies modernization opportunities, and provides step-by-step migration recommendations. The system should understand code dependencies, assess migration risks, and generate modernized code while preserving business logic.

Steps:

- Design code analysis engines for legacy language understanding and dependency mapping
- Implement modernization strategy generation based on current technology stacks
- Create risk assessment frameworks for migration planning and execution
- Build code transformation tools with business logic preservation
- Develop testing strategy generation for validating modernized systems
- Include project planning and resource estimation for modernization efforts

Suggested Data Requirements:

- Legacy codebase samples across different languages and technologies
- Modernization case studies and transformation examples
- Technology migration patterns and best practices
- Risk assessment criteria and mitigation strategies

Themes: AI in software engineering lifecycle, AI for Reengineering

The steps and data requirements outlined above are intended solely as reference points to assist you in conceptualising your solution.

54. AI-Powered Personalized Medicine Platform

Summary: Build a comprehensive platform that integrates genomic data, medical records, and lifestyle information to provide personalized treatment recommendations and drug discovery insights.

Problem Statement: Personalized medicine requires integration of complex multi-modal data to provide tailored treatment recommendations. Your task is to create a platform that combines genomic sequencing data, electronic health records, lifestyle factors, and medical literature to suggest personalized treatment plans. The system should predict drug responses, identify potential adverse reactions, and support precision medicine while ensuring privacy and regulatory compliance.

Steps:

- Design multi-modal data integration for genomic, clinical, and lifestyle data
- Implement drug response prediction models based on genetic markers and patient profiles
- Create adverse reaction prediction and drug interaction analysis
- Build treatment recommendation engines using evidence-based medicine
- Develop clinical trial matching and patient stratification capabilities
- Include privacy-preserving analytics and regulatory compliance frameworks

Suggested Data Requirements:

- De-identified genomic and clinical datasets
- Drug response and adverse reaction databases
- Medical literature and clinical guidelines
- Lifestyle and environmental factor data

Themes: AI for Industry, Responsible AI

The steps and data requirements outlined above are intended solely as reference points to assist you in conceptualising your solution.

55. Climate Change Impact Modeling and Mitigation Platform

Summary: Create an AI platform that models climate change impacts, predicts environmental changes, and recommends mitigation strategies for organizations and governments.

Problem Statement: Climate change requires sophisticated modeling and strategic planning for effective mitigation and adaptation. Your task is to build a platform that integrates climate data, economic models, and environmental sensors to predict climate impacts and recommend mitigation strategies. The system should model various scenarios, assess risks, and provide actionable recommendations for different stakeholders while supporting policy decision-making.

Steps:

- Design climate modeling integration combining multiple data sources and prediction models
- Implement scenario analysis for different climate change and mitigation pathways
- Create economic impact assessment and cost-benefit analysis tools
- Build risk assessment frameworks for different sectors and regions
- Develop mitigation strategy recommendation engines with feasibility analysis
- Include policy simulation and stakeholder impact assessment capabilities

Suggested Data Requirements:

- Climate and weather datasets with historical and projection data
- Economic and demographic data for impact assessment
- Environmental sensor data and satellite imagery
- Mitigation strategy case studies and effectiveness data

Themes: Classical AI/ML/DL for prediction, Deep-tech research

The steps and data requirements outlined above are intended solely as reference points to assist you in conceptualising your solution.

56. Universal Language Translation and Communication Platform

Summary: Develop a real-time, multi-modal translation platform that handles text, speech, and visual content across hundreds of languages including low-resource languages.

Problem Statement: Global communication requires sophisticated translation capabilities that go beyond text to include cultural context, visual elements, and real-time conversation. Your task is to create a universal translation platform that handles multiple modalities, preserves cultural nuances, and supports low-resource languages. The system should provide real-time translation, cultural adaptation, and context-aware communication while maintaining accuracy and cultural sensitivity.

Steps:

- Design multi-modal translation combining text, speech, and visual content understanding

- Implement low-resource language support using transfer learning and data augmentation
- Create cultural context preservation and adaptation mechanisms
- Build real-time conversation translation with context maintenance
- Develop visual content translation including signs, documents, and cultural symbols
- Include quality assessment and cultural sensitivity validation frameworks

Suggested Data Requirements:

- Parallel text corpora across multiple language pairs
- Speech datasets with cultural and dialectal variations
- Visual content with multilingual text and cultural symbols
- Cultural context and adaptation guidelines

Themes: Multi-modal UX, GenAI & its techniques

The steps and data requirements outlined above are intended solely as reference points to assist you in conceptualising your solution.